VLAD THE STAKER

~ or ~

how to stop burning

HISTORY: TOO MUCH VOLATILITY

GOODTIMES was a tiny XGBoost model with a hillclimbing process to pick what eras to train on.

- Correlation scores were calculated per era,
- Fitness function was optimizing the mean of the top-scoring eras

- It was a caricature of the average XGBoost model, more extreme because it wasn't dampened by the "burn" eras

- While GOODTIMES performed better on average than my other models, I really didn't like the volatility!

PAY ATTENTION TO WHAT YOU STAKE

- Richard Craib at some point made a comment that stuck with me:

people don’t spend enough time thinking about stake allocation

STAKE ALLOCATION == STOCK ALLOCATION

If you had to describe the tournament, in terms of finance:

1. Stock allocation, long only.
2. Two markets (CORR and TC)
3. Leverage allowed at 0.5x or 1x for CORR
4. Leverage allowed at 0.5x-3x for TC

STAKE ALLOCATION == STOCK ALLOCATION

Some iterations I went through:

1. Pick 4-5 models that I like, and brute-force NMR allocation and CORR and TC multipliers to get a higher sharpe.

2. Use the R package fPortfolio and "robust" time series generation based on covariance matrices

3. Use the R package fPortfolio and some resampling

WITHIN FPORTFOLIO – COMPLEX

- fPortfolio wants a complete timeseries
- I continuously add models

Solution: estimate the covariance matrix between models and build a virtual time series from that

WITHIN FPORTFOLIO – COMPLEX

- *"Robust covariance matrix estimation"* still had a ton of demands on how complete your time series had to be.
- To get more data per model, I was using daily interim performances as well
- These robust methods never felt particularly robust.

WITHIN FPORTFOLIO – RESAMPLE

- With the daily tournament, you get correlated, but more frequent ”final” scores of your model

- No longer considering any time series estimations. Just using the subset of performances that overlap between the models

So, in more detail, what is Vlad doing?

VLAD THE STAKER – DATA INPUT

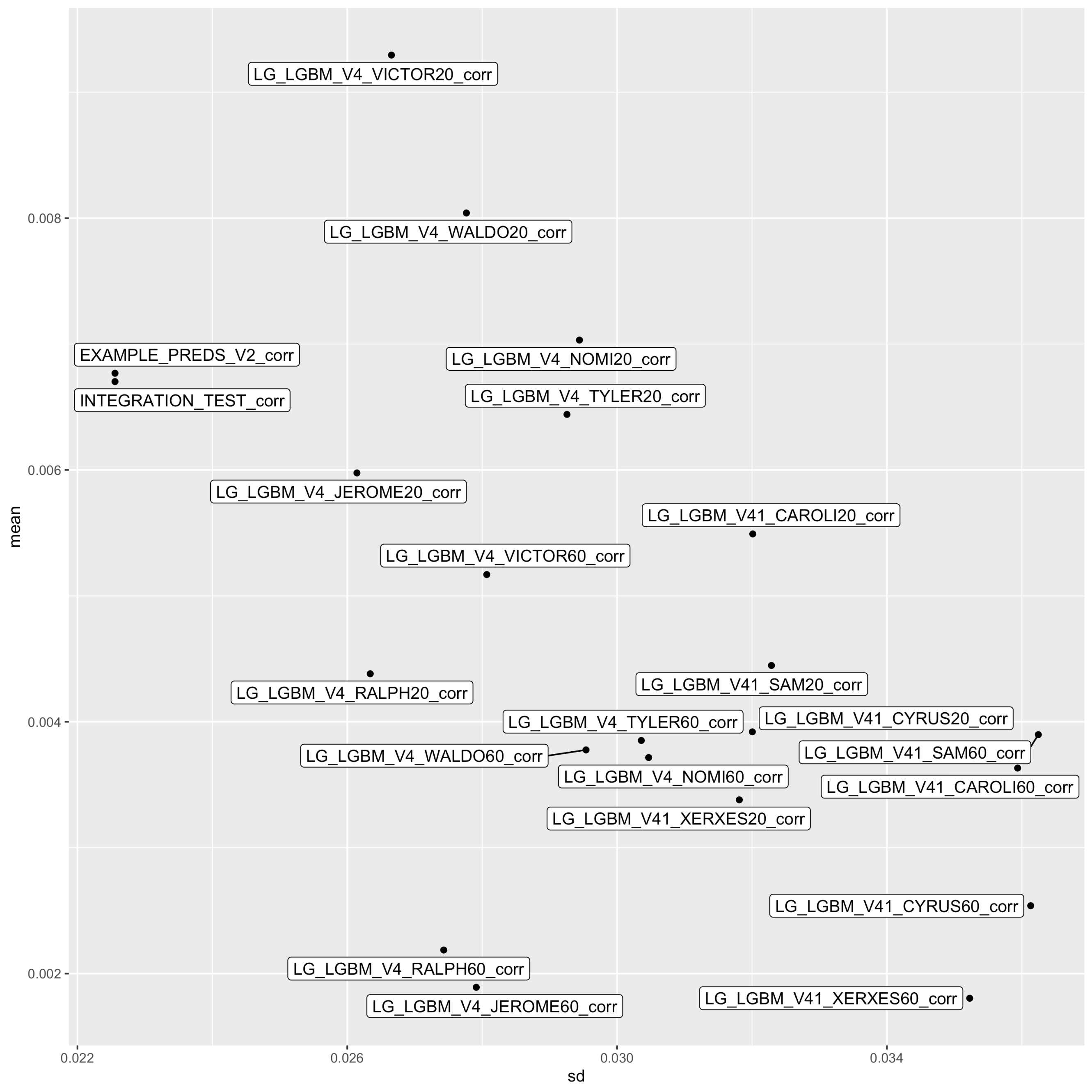
- Vlad reads in an excel sheet with the models you want to have included in your stake optimization, including a starting and ending era.

- Here the starting era is also the start of the daily tournament

	A	B	C	
1	ModelName	Starting Era	Last Era	Notes
2	INTEGRATION_TEST	339	1000	-
3	EXAMPLE_PREDS_V2	339	1000	-
4	LG_LGBM_V4_NOMI20	339	1000	-
5	LG_LGBM_V4_RALPH20	339	1000	-
6	LG_LGBM_V4_TYLER20	339	1000	-
7	LG_LGBM_V4_WALDO20	339	1000	-
8	LG_LGBM_V4_JEROME20	339	1000	-
9	LG_LGBM_V4_VICTOR20	339	1000	-
10	LG_LGBM_V4_NOMI60	339	1000	-
11	LG_LGBM_V4_RALPH60	339	1000	-
12	LG_LGBM_V4_TYLER60	339	1000	-
13	LG_LGBM_V4_WALDO60	339	1000	-
14	LG_LGBM_V4_JEROME60	339	1000	-
15	LG_LGBM_V4_VICTOR60	339	1000	-
16	LG_LGBM_V41_CYRUS20	339	1000	-
17	LG_LGBM_V41_CAROLI20	339	1000	-
18	LG_LGBM_V41_XERXES20	339	1000	-

MODEL PERF.

- Vlad splits models up into _corr and _tc variants. That makes everything easier for stake weighting.
- For corr and TC, Vlad plots the mean score per model and standard deviation



STEP 1. SUBSETTING

- Vlad only considers the subset of models that is on average positive for corr or positive for TC.

- Vlad filters the dataset for days where we have information on all models

Get at least 60 resolved days of scores.

STEP 2. RESAMPLING

I want stability. I don't want to drastically redistribute my stakes every week (when stake management?)

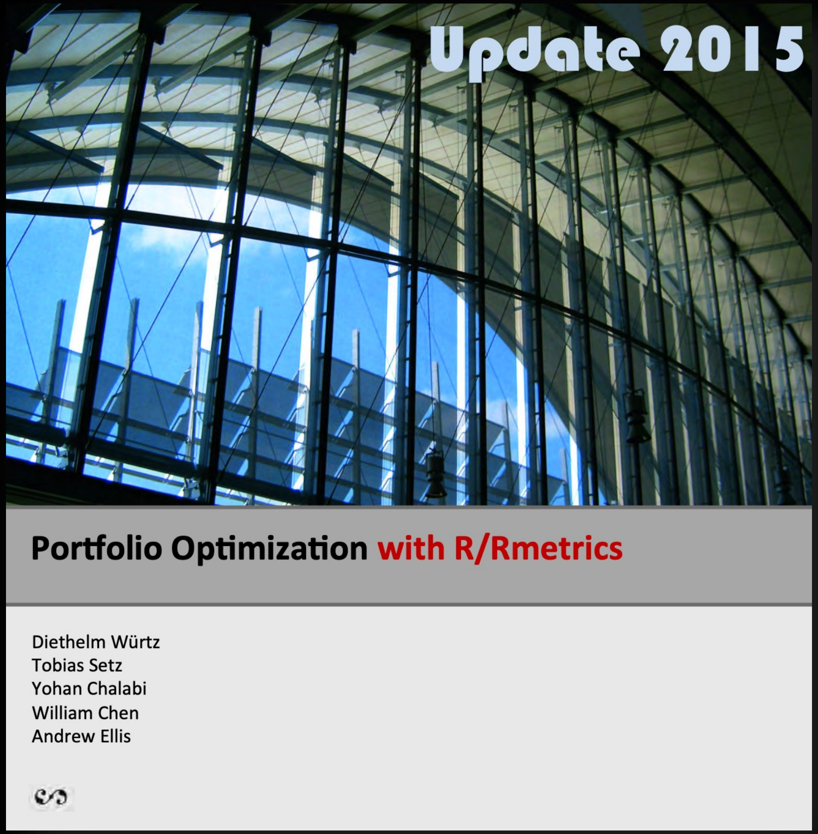
- Longer time series work better
- But resampling the time series you have 40 times works as well

Vlad resamples

STEP 3. PICK A PORTFOLIO STRATEGY

There are a bunch of different goals you can specify when optimizing portfolios.

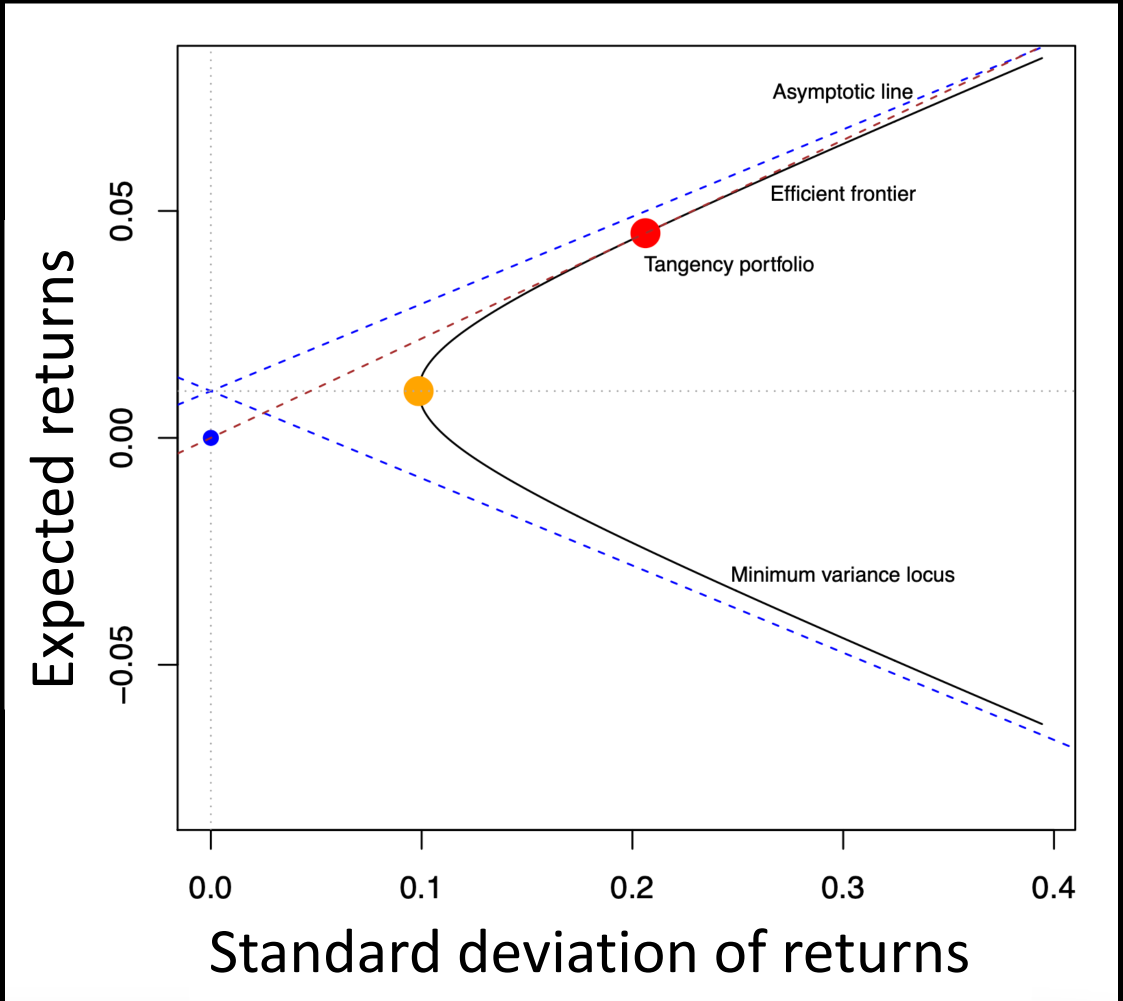
(and this is not my field at all)



STEP 3. PICK A PORTFOLIO STRATEGY

Vlad averages between the red dot (tangency portfolio), and the orange dot (minvariance portfolio)

Both are points on the efficient frontier - a theoretical description of the tradeoff between higher returns and higher variance



VLAD OUTPUT – TABLE 1

- All 22 Numer.ai example models had a positive CORR and 10 models had a positive TC.

- Only 4 made it through the selection to be stake-weighted

Table: Suggested portfolio. For models with both a _corr and a _tc component, either set up two model slots, or find a multiplier that works for both stakes

name	corr_weight	tc_weight	corr_0.5x	corr_1x	tc_0.5x	tc_1x	tc_1.5x	tc_2x	tc_2.5x	tc_3x
INTEGRATION_TEST	0.380	0.095	760	380	190	100	60	50	40	30
LG_LGBM_V4_JEROME20	0.000	0.211	0	0	420	210	140	110	80	70
LG_LGBM_V4_TYLER20	0.000	0.055	0	0	110	60	40	30	20	20
LG_LGBM_V4_VICTOR20	0.249	0.000	500	250	0	0	0	0	0	0

VLAD OUTPUT – TABLE 1

- INTEGRATION_TEST and VICTOR20 are CORR worthy
- INTEGRATION_TEST, JEROME20 and TYLER20 are TC worthy

So you need 5 stakeslots

Table: Suggested portfolio. For models with both a _corr and a _tc component, either set up two model slots, or find a multiplier that works for both stakes

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LG_LGBM_V4_TYLER20	0.000	0.055	0	0	110	60	40	30	20	20
LG_LGBM_V4_VICTOR20	0.249	0.000	500	250	0	0	0	0	0	0

VLAD OUTPUT – TABLE 2

Table 2 shows the performance characteristics of Vlad's portfolio

compared to: (1) tangency portfolio, (2) minvariance portfolio, (3) portfolios based on the individual models.

Table: Expected returns based on separated weights

	mean	Cov	CVaR	VaR	samplesize
portfolio	0.0070	0.0205	0.0330	0.0287	178
portfolio1_tangency	0.0076	0.0216	0.0347	0.0306	178
portfolio2_minvariance	0.0064	0.0203	0.0322	0.0282	178
INTEGRATION_TEST	0.0030	0.0098	0.0176	0.0144	178
LG_LGBM_V4_JEROME20	0.0014	0.0065	0.0109	0.0088	178
LG_LGBM_V4_TYLER20	0.0004	0.0019	0.0026	0.0022	178
LG_LGBM_V4_VICTOR20	0.0022	0.0066	0.0112	0.0106	178

VLAD OUTPUT – TABLE 2

Table 2 shows the performance characteristics of Vlad's portfolio

compared to: (1) tangency portfolio, (2) minvariance portfolio, (3) portfolios based on the individual models.

There is still quite some risk here.

But these are all models which don't consider per-era performance. Just the average.

Table: Expected returns based on separated weights

	mean	Cov	CVaR	VaR	samplesize
portfolio	0.0070	0.0205	0.0330	0.0287	178
portfolio1_tangency	0.0076	0.0216	0.0347	0.0306	178
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VLAD CHURN

Vlad's suggestions change over time, as more scores are added.

Changes:

- *Vlad added victor20 (25% of portfolio)*
- Vlad dropped ralph20 (10%) and waldo20 (7%)
- Integration_test stake weight went from 16% CORR and 17% TC to 38% CORR and 10% TC

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Table: Expected returns based on separated weights

	mean	Cov	CVaR	VaR	samplesize
portfolio	0.0094	0.0238	0.0375	0.0313	140
portfolio1_tangency	0.0102	0.0252	0.0387	0.0307	140
portfolio2_minvariance	0.0085	0.0235	0.0367	0.0313	140
INTEGRATION_TEST	0.0032	0.0078	0.0147	0.0108	140
LG_LGBM_V4_JEROME20	0.0013	0.0042	0.0068	0.0061	140
LG_LGBM_V4_RALPH20	0.0006	0.0025	0.0041	0.0039	140
LG_LGBM_V4_TYLER20	0.0036	0.0122	0.0153	0.0138	140
LG_LGBM_V4_WALDO20	0.0007	0.0019	0.0032	0.0031	140

VLAD OUTPUT

My own models don't do as well as the example models, but they do have less risk

(comparison is not fully ==as I didn't start daily submissions from the start, but time period is equal)

Table: Expected returns based on separated weights

	mean	Cov	CVaR	VaR	samplesize
portfolio	0.0070	0.0205	0.0330	0.0287	178
portfolio1_tangency	0.0076	0.0216	0.0347	0.0306	178
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Table: Expected returns based on separated weights

	mean	Cov	CVaR	VaR	samplesize
portfolio	0.0044	0.0118	0.0160	0.0150	137
portfolio1_tangency	0.0046	0.0122	0.0166	0.0152	137
portfolio2_minvariance	0.0042	0.0114	0.0159	0.0149	137
bor_darwin_fnh07_en	0.0024	0.0073	0.0131	0.0093	137
bor_evolvedape	0.0007	0.0034	0.0076	0.0047	137
bor_lamarck_fnh05_en	0.0002	0.0013	0.0024	0.0020	137
bor_wallace_fns08	0.0006	0.0018	0.0034	0.0029	137
bor_wallace_fnws11	0.0006	0.0023	0.0043	0.0035	137

VLAD IS WAITING FOR YOU

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